AIDCF

AI Data Centre Facilities Assurance Framework

REFERENCE IMPLEMENTATION, SYNTHETIC DATASET AND CONFORMANCE PILOT

Version 1.0

A bounded, non-production demonstration of AIDCF architecture, machine-readable records, traceability, controls, evidence, assessment, gates, decisions, reports, events and AITG handoff

Artifact ID: AIDCF-RIP-001
Status: Draft for controlled review
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Depends on: AIDCF-003 through AIDCF-014
Dataset status: Entirely synthetic and illustrative

Document control

Reference implementation requirement.

Element	Pilot specification
Purpose	Demonstrate that AIDCF can be instantiated, validated, traced, assessed, decided, reported and reassessed end to end.
Pilot subject	Synthetic high-density liquid-cooled distributed AI training capability.
Outputs	Baseline manifest, registries, schemas, semantic artifacts, synthetic records, tests, traceability graph, assessment, decision, reports and event-driven reassessment.
Non-production	No record describes a real facility, provider, product deployment, engineering design or assurance conclusion.
Safety	No operational switching, intrusive testing or hazardous action is authorised.
Non-certification	Technical conformance and synthetic pilot completion do not create AIDCF certification or accreditation.

Executive thesis

AIDCF-015 proves the framework by exercising one bounded synthetic capability from architecture definition through machine validation, assurance evaluation, reporting and change impact. The pilot is designed to reveal gaps, contradictions and implementation burden before platform development.

Pilot chain

Baseline → Synthetic Capability → Architecture → Dependencies → Scenarios → Safeguards → Controls → Evidence → Assessment → Findings/UNKNOWN → Gates → Scores → Decision → Reports → Change Event → Impact Analysis → Reassessment → Supersession → AITG Handoff

1 Pilot principles

Reference implementation requirement.

Element	Pilot specification
Synthetic by construction	Required
Capability first	Required
No hidden normalisation	Required
UNKNOWN preserved	Required
Gate dominance	Required
Scores contextual only	Required
Human authority	Required
Immutable snapshots	Required
Reverse traceability	Required
Security classification	Required
Reproducible validation	Required
No certification	Required

2 Success criteria

Reference implementation requirement.

Element	Pilot specification
Baseline issues visible	Required
Canonical IDs resolve	Required
All required records validate	Required
Forward trace complete	Required
Reverse trace complete	Required
At least one material UNKNOWN	Required
At least one failed or conditional fixture	Required
Gate constraints operate	Required
Score is reproducible	Required
Report matches decision	Required
Change impact identifies affected records	Required
AITG handoff preserves ownership	Required

3 Out of scope

Reference implementation requirement.

Element	Pilot specification
Real engineering design	Required
Vendor selection	Required
Live telemetry ingestion	Required
Operational testing	Required
Legal or regulatory opinion	Required
Production security accreditation	Required
Real customer attestation	Required
Performance benchmarking	Required
Automated acceptance of risk	Required
Production platform SLA	Required

4 Conformance target

Reference implementation requirement.

Element	Pilot specification
AIDCF-014 T0 vocabulary	Required
T1 records	Required
T2 traceability	Required
T3 JSON validation	Required
T4 graph validation	Required
T5 decision-ready	Required
T6 event and reassessment	Required
T7 fixture-tested technical conformance	Required

5 Pilot governance

Reference implementation requirement.

Element	Pilot specification
Pilot sponsor	Required
Methodology authority	Required
Pilot lead	Required
Architecture lead	Required
Domain specialists	Required
Evidence lead	Required
Data/schema lead	Required
Independent reviewer	Required
Decision authority	Required
Security reviewer	Required
AITG reviewer	Required
Release manager	Required

6 Baseline manifest

Reference implementation requirement.

Element	Pilot specification
releaseId	aidcf:release:pilot-015-v1.0
Status	CONDITIONALLY_BASELINED
Included artifacts	AIDCF-003 through AIDCF-014
Known blocker	AIDCF-007 has conflicting published control-universe counts.
Pilot treatment	Use only explicitly selected representative controls; do not claim catalogue completeness.
Resolution requirement	Authoritative correction and migration before public baseline.
Integrity	Manifest and registry hashes recorded in implementation package.
Approval	Methodology and independent review required.

7 BaselineIssue record

Reference implementation requirement.

Element	Pilot specification
issueId	aidcf:baseline-issue:control-cardinality-001
Sources	AIDCF-007 cover and document-control section
Conflict	15/90/360 versus 14/84/336.
Impact	Control coverage, registry completeness, scoring denominator and release claims.
Pilot disposition	Open; does not block selected-control pilot.
Prohibited claim	No assertion that the pilot covers the full control universe.
Resolution owner	Methodology authority.
Migration	Reissue affected registry and downstream cardinality declarations.

8 Synthetic capability record

Reference implementation requirement.

Element	Pilot specification
capabilityId	aidcf:capability:synthetic-training-001
Name	Synthetic Liquid-Cooled Distributed Training Capability
Purpose	Execute bounded distributed model-training jobs using a synthetic high-density data-centre architecture.
Workload	Distributed training with checkpoint and restart.
Boundary	Synthetic site, facility systems, AI cluster, storage, fabrics, operations and named external services.
Lifecycle	Operational-assurance pilot snapshot.
Criticality	Illustrative high consequence for completion, integrity, thermal stability and controlled recovery.
Owner	Synthetic Capability Owner role.
Classification	INTERNAL / SYNTHETIC.
Decision question	Can the synthetic capability support the declared outcomes within the modelled states and evidence limitations?

9 Synthetic Service Outcomes

Reference implementation requirement.

Outcome	Name	Synthetic statement
SO-01	Training service availability	Service remains available within declared normal/degraded conditions.
SO-02	Distributed job completion	Jobs complete or fail in a controlled, observable manner.
SO-03	Checkpoint integrity and recoverability	Checkpoints are complete, consistent and restorable.
SO-04	Thermal stability	Rack and accelerator thermal conditions remain within declared synthetic envelope.
SO-05	Controlled recovery	Defined failures are contained and recovery is validated before return.

10 Required functions

Reference implementation requirement.

ID	Required function
FN-01	Accept and schedule workload
FN-02	Provide accelerator compute
FN-03	Move distributed-training traffic
FN-04	Store datasets/checkpoints
FN-05	Deliver electrical power
FN-06	Remove heat through liquid cooling
FN-07	Monitor and control infrastructure
FN-08	Detect/contain incident
FN-09	Restore training service
FN-10	Preserve evidence and decision lineage

11 Synthetic system inventory

Reference implementation requirement.

ID	System	Domain
SYS-PWR-01	Incoming power	POW
SYS-PWR-02	UPS and battery	POW
SYS-PWR-03	Generators and fuel	POW
SYS-PWR-04	Distribution and busway	POW
SYS-THM-01	Primary cooling plant	THM
SYS-THM-02	CDU Train A	THM
SYS-THM-03	CDU Train B	THM
SYS-THM-04	Liquid loops/manifolds	THM
SYS-CMP-01	Accelerator cluster	CMP
SYS-STO-01	Dataset storage	STO
SYS-STO-02	Checkpoint service	STO
SYS-NET-01	Scale-out fabric	NET
SYS-NET-02	Storage fabric	NET
SYS-SEC-01	Management identity/control plane	SEC
SYS-OPS-01	Operations centre	OPS
SYS-EVA-01	Evidence and assessment repository	EVA

12 Representative asset instances

Reference implementation requirement.

ID	Synthetic asset	System
AST-001	Utility service point	SYS-PWR-01
AST-002	UPS A	SYS-PWR-02
AST-003	UPS B	SYS-PWR-02
AST-004	Generator A	SYS-PWR-03
AST-005	Generator B	SYS-PWR-03
AST-006	CDU A controller	SYS-THM-02
AST-007	CDU B controller	SYS-THM-03
AST-008	Shared control-network switch	SYS-SEC-01
AST-009	Accelerator rack group	SYS-CMP-01
AST-010	Checkpoint metadata service	SYS-STO-02
AST-011	Fabric spine group	SYS-NET-01
AST-012	Evidence store	SYS-EVA-01

13 Interface contracts

Reference implementation requirement.

ID	Interface	Type
IF-PWR-01	Utility to switchgear	Electrical
IF-PWR-02	UPS to busway	Electrical
IF-THM-01	Plant to CDU A	Thermal/fluid
IF-THM-02	Plant to CDU B	Thermal/fluid
IF-THM-03	CDUs to rack manifolds	Thermal/fluid
IF-NET-01	Compute to scale-out fabric	Network
IF-STO-01	Compute to checkpoint service	Data/network
IF-CTL-01	Management identity to BMS/EPMS/CDUs	Management
IF-EVA-01	Sources to evidence repository	Assurance

14 Dependency register

Reference implementation requirement.

ID	Source	Target	Criticality
DEP-01	Accelerator cluster	Electrical distribution	Critical
DEP-02	Accelerator cluster	CDUs and liquid loops	Critical
DEP-03	Training job	Scale-out fabric	Critical
DEP-04	Job recovery	Checkpoint service	Critical
DEP-05	Both CDU trains	Shared control-network switch	Common-mode critical
DEP-06	Generator continuity	Fuel replenishment service	External critical
DEP-07	Operations	Management identity service	Critical
DEP-08	Decision lineage	Evidence repository	Critical
DEP-09	Recovery	OEM support and spares	External material

15 Fault-domain register

Reference implementation requirement.

ID	Fault domain	Potential scope
FD-01	Utility supply boundary	Incoming power dependencies
FD-02	Electrical room/common distribution	UPS and downstream paths
FD-03	Cooling control network	Both CDU trains
FD-04	Cooling plant common header	Both liquid-cooling trains
FD-05	Training fabric control plane	Distributed job communication
FD-06	Checkpoint metadata domain	Checkpoint discovery and restore
FD-07	Management identity domain	Privileged facility and platform operations

16 Operating-state model

Reference implementation requirement.

State	Synthetic meaning
NORMAL	All declared systems available.
MAINTENANCE	One synthetic redundant train intentionally unavailable.
DEGRADED	Reduced redundancy/capacity; workload restrictions apply.
EMERGENCY	Protective response and workload containment.
RECOVERY	Restoration and validation in progress.
TRANSITION	Failover, restart or controlled return.
UNKNOWN	Reliable evidence cannot establish state.

17 Capacity assertions

Reference implementation requirement.

ID	Assertion	Representation	Limitation
CAP-01	Accelerator demand	Synthetic units only	No real engineering value
CAP-02	Electrical available capacity	Qualitative sufficient/limited/unknown	Synthetic evidence
CAP-03	Thermal available capacity	Qualitative sufficient/limited/unknown	Synthetic evidence
CAP-04	Checkpoint performance envelope	Qualitative normal/degraded	Synthetic evidence
CAP-05	Recovery capacity	Reserved/not proven	UNKNOWN retained

18 Failure-scenario catalogue

Reference implementation requirement.

ID	Scenario	Initiating condition	Potential outcome effect
SCN-01	CDU A loss under maintenance exposure	CDU A fails while CDU B is carrying declared load.	Thermal degradation; workload throttling; possible controlled stop.
SCN-02	Shared cooling control-network failure	Shared switch prevents control/telemetry across both CDU trains.	Common-mode loss of visibility/control; emergency containment.
SCN-03	Checkpoint metadata degradation	Metadata service becomes unavailable during checkpoint commit.	Checkpoint status becomes UNKNOWN; restart claim constrained.
SCN-04	Utility loss with Generator B unavailable	Utility loss coincides with generator maintenance.	Reduced continuity margin and conditional runtime.
SCN-05	Privileged automation error	Authorised automation issues incorrect cooling setpoint.	Thermal excursion risk and evidence/authority challenge.

Scenario workbook: SCN-01

Reference implementation requirement.

Element	Pilot specification
Name	CDU A loss under maintenance exposure
Pre-state	NORMAL or declared maintenance state.
Initiating condition	CDU A fails while CDU B is carrying declared load.
Propagation	Explicit system/dependency/fault-domain steps.
Safeguards	Detection, containment, workload control and recovery measures.
Residual state	Thermal degradation; workload throttling; possible controlled stop.
Evidence	Synthetic telemetry, configuration, procedure and test records.
UNKNOWN	Any unverified independence, restoration or source-health fact.
Gates	Applicable CG results and decision effect.
Review	Technical and independent review required.

Scenario workbook: SCN-02

Reference implementation requirement.

Element	Pilot specification
Name	Shared cooling control-network failure
Pre-state	NORMAL or declared maintenance state.
Initiating condition	Shared switch prevents control/telemetry across both CDU trains.
Propagation	Explicit system/dependency/fault-domain steps.
Safeguards	Detection, containment, workload control and recovery measures.
Residual state	Common-mode loss of visibility/control; emergency containment.
Evidence	Synthetic telemetry, configuration, procedure and test records.
UNKNOWN	Any unverified independence, restoration or source-health fact.
Gates	Applicable CG results and decision effect.
Review	Technical and independent review required.

Scenario workbook: SCN-03

Reference implementation requirement.

Element	Pilot specification
Name	Checkpoint metadata degradation
Pre-state	NORMAL or declared maintenance state.
Initiating condition	Metadata service becomes unavailable during checkpoint commit.
Propagation	Explicit system/dependency/fault-domain steps.
Safeguards	Detection, containment, workload control and recovery measures.
Residual state	Checkpoint status becomes UNKNOWN; restart claim constrained.
Evidence	Synthetic telemetry, configuration, procedure and test records.
UNKNOWN	Any unverified independence, restoration or source-health fact.
Gates	Applicable CG results and decision effect.
Review	Technical and independent review required.

Scenario workbook: SCN-04

Reference implementation requirement.

Element	Pilot specification
Name	Utility loss with Generator B unavailable
Pre-state	NORMAL or declared maintenance state.
Initiating condition	Utility loss coincides with generator maintenance.
Propagation	Explicit system/dependency/fault-domain steps.
Safeguards	Detection, containment, workload control and recovery measures.
Residual state	Reduced continuity margin and conditional runtime.
Evidence	Synthetic telemetry, configuration, procedure and test records.
UNKNOWN	Any unverified independence, restoration or source-health fact.
Gates	Applicable CG results and decision effect.
Review	Technical and independent review required.

Scenario workbook: SCN-05

Reference implementation requirement.

Element	Pilot specification
Name	Privileged automation error
Pre-state	NORMAL or declared maintenance state.
Initiating condition	Authorised automation issues incorrect cooling setpoint.
Propagation	Explicit system/dependency/fault-domain steps.
Safeguards	Detection, containment, workload control and recovery measures.
Residual state	Thermal excursion risk and evidence/authority challenge.
Evidence	Synthetic telemetry, configuration, procedure and test records.
UNKNOWN	Any unverified independence, restoration or source-health fact.
Gates	Applicable CG results and decision effect.
Review	Technical and independent review required.

24 Safeguard register

Reference implementation requirement.

ID	Safeguard	Function
SG-01	Power-path monitoring	Detect
SG-02	UPS/generator transfer sequence	Prevent/Recover
SG-03	Dual CDU architecture	Prevent/Contain
SG-04	Independent local thermal protection	Contain
SG-05	Workload throttling/controlled stop	Contain
SG-06	Checkpoint validation	Detect
SG-07	Checkpoint restore test	Recover
SG-08	Privileged-access control	Prevent
SG-09	Command logging and rollback	Detect/Correct
SG-10	Incident and recovery procedure	Respond/Recover
SG-11	Evidence integrity and review	Assure

25 Representative control profile

Reference implementation requirement.

Pilot control	Domain	Pilot objective
CTL-001	CAP	Capability and outcomes defined
CTL-002	DEP	Critical dependencies visible
CTL-003	POW	Power continuity design and operating evidence
CTL-004	THM	Thermal continuity and liquid-cooling evidence
CTL-005	CMP	Compute configuration and health
CTL-006	STO	Checkpoint integrity and restore
CTL-007	NET	Fabric availability and monitoring
CTL-008	OPS	Work control, incident and change management
CTL-009	RES	Containment and recovery validation
CTL-010	SEC	Privileged digital/OT authority
CTL-011	EVA	Evidence provenance and quality
CTL-012	EVA	Gate/decision/reassessment integrity

26 Control coverage limitation

Reference implementation requirement.

Element	Pilot specification
Important	Pilot control IDs are local synthetic identifiers, not replacements for canonical MCF IDs.
Reason	AIDCF-007 cardinality and catalogue baseline must be harmonised before authoritative full-control mapping.
Permitted use	Demonstrate traceability, evidence, assessment, gates and decision logic.
Prohibited use	Claim full AIDCF-007 control coverage or real-world conformity.
Migration	Replace local IDs with approved canonical control registry after baseline release.

27 Evidence-object register

Reference implementation requirement.

ID	Evidence	Class	Status
EV-001	Synthetic architecture snapshot	Documentary	Accepted
EV-002	Synthetic power single-line assurance view	Documentary	Accepted
EV-003	Synthetic cooling P&ID assurance view	Documentary	Accepted
EV-004	Synthetic power telemetry extract	Operational	Accepted
EV-005	Synthetic cooling telemetry extract	Operational	Accepted
EV-006	Synthetic compute/fabric health extract	Operational	Accepted
EV-007	Synthetic checkpoint integrity record	Test	Accepted
EV-008	Synthetic restore exercise	Test	Qualified
EV-009	Synthetic privileged-access review	Configuration	Accepted
EV-010	Synthetic incident/recovery drill	Exercise	Accepted
EV-011	CDU control-network independence evidence	Expected	Missing/UNKNOWN
EV-012	Synthetic evidence manifest/hash register	Integrity	Accepted

28 Evidence quality ratings

Reference implementation requirement.

Element	Pilot specification
Provenance	Declared synthetic source and creator.
Freshness	Aligned to pilot snapshot.
Coverage	Representative, not complete.
Reliability	Internally consistent synthetic fixture.
Traceability	Linked to assertions, controls and gates.
Integrity	Hash field and immutable snapshot illustrated.
Sufficiency	Claim-specific; EV-011 intentionally insufficient.
Restriction	Synthetic INTERNAL classification.

29 Assessment procedures

Reference implementation requirement.

ID	Procedure
AP-01	Review capability/outcome records
AP-02	Reconcile architecture and system inventory
AP-03	Trace dependencies and fault domains
AP-04	Review scenario propagation
AP-05	Inspect synthetic evidence manifest
AP-06	Reperform selected trace links
AP-07	Evaluate representative control results
AP-08	Evaluate gates
AP-09	Reproduce score snapshot
AP-10	Challenge decision/report consistency
AP-11	Inject change event and run impact analysis
AP-12	Validate AITG handoff

30 Synthetic control results

Reference implementation requirement.

Control	Result	Confidence
CTL-001	SUPPORTED	High
CTL-002	SUPPORTED	Medium
CTL-003	SUPPORTED	Medium
CTL-004	PARTIAL	Medium
CTL-005	SUPPORTED	Medium
CTL-006	PARTIAL	Medium
CTL-007	SUPPORTED	Medium
CTL-008	SUPPORTED	Medium
CTL-009	PARTIAL	Medium
CTL-010	SUPPORTED	Medium
CTL-011	SUPPORTED	High
CTL-012	SUPPORTED	Medium

31 Finding register

Reference implementation requirement.

ID	Class	Condition	Status
FND-001	Significant weakness	Shared CDU control-network independence is not evidenced.	Open
FND-002	Weakness	Checkpoint restore evidence is qualified and does not cover every declared recovery condition.	Open
FND-003	Observation	Generator maintenance state reduces synthetic continuity margin.	Open

32 UNKNOWN register

Reference implementation requirement.

ID	Unknown	Materiality	Decision effect
UNK-001	Independence of both CDU trains from shared control-network failure	Critical	Blocks unqualified thermal resilience conclusion
UNK-002	Recovery capacity under simultaneous power and cooling degradation	Material	Limits recovery assurance
UNK-003	OEM support availability during synthetic recovery window	Moderate	Condition on recovery decision

33 Gate-results register

Reference implementation requirement.

Gate	Status	Pilot basis
CG-01	PASS	Capability/outcomes defined
CG-02	PASS	Boundary and operating envelope defined for pilot
CG-03	CONDITIONAL	External dependencies modelled with limitations
CG-04	PASS	Safety boundary and synthetic-only testing
CG-05	CONDITIONAL	Power continuity supported with maintenance limitation
CG-06	UNKNOWN	Cooling-control common mode unresolved
CG-07	PASS	Control-system authority modelled
CG-08	PASS	Compute/storage/network integrity represented
CG-09	UNKNOWN	Containment under common-mode cooling failure unresolved
CG-10	CONDITIONAL	Recovery evidence qualified
CG-11	PASS	Evidence manifest and lineage validated
CG-12	PASS	Critical UNKNOWN visible
CG-13	PASS	Roles and decisions defined
CG-14	PASS	Change control and review defined
CG-15	PASS	Digital authority path represented

34 Score snapshot

Reference implementation requirement.

Element	Pilot specification
Profile	Synthetic Pilot Profile v1.0
Controls assessed	12 representative synthetic controls
Raw indicator	Illustrative only; not a framework-wide score.
Coverage	Representative pilot coverage, not full catalogue coverage.
Confidence	Medium overall due to synthetic evidence and open UNKNOWN.
Gate overlay	CG-06 and CG-09 UNKNOWN control the decision ceiling.
Anti-gaming	UNKNOWN excluded from favourable averaging and remains explicit.
Reproducibility	Inputs, weights and calculations fixed in snapshot.
Prohibited interpretation	No facility rating, certification, benchmark or real assurance.

35 Decision recommendation

Reference implementation requirement.

Element	Pilot specification
Recommended state	LIMITED
Reason 1	Pilot architecture, evidence and traceability are substantially complete for demonstration.
Reason 2	Critical cooling-control common-mode independence remains UNKNOWN.
Reason 3	Recovery evidence is qualified.
Controlling gates	CG-06 UNKNOWN; CG-09 UNKNOWN; CG-10 CONDITIONAL.
Permitted reliance	Technical demonstration of AIDCF workflow.
Prohibited reliance	Real facility, safety, resilience or certification conclusion.
Conditions	Resolve UNK-001 and broaden recovery evidence before stronger synthetic conclusion.

36 Synthetic assurance decision

Reference implementation requirement.

Element	Pilot specification
decisionId	aidcf:decision:synthetic-training-001
Decision	LIMITED
Authority	Synthetic Decision Authority role.
Scope	Defined synthetic capability and pilot snapshot only.
Evidence period	Pilot fixture snapshot.
Conditions	Open findings/UNKNOWN and gate constraints.
Validity	Until synthetic change event or superseding snapshot.
Non-certification	Not a certificate, guarantee or engineering approval.

37 Executive report example

Reference implementation requirement.

Element	Pilot specification
Decision	LIMITED
What is supported	End-to-end traceability and machine validation operate on the synthetic dataset.
Top risk	Shared cooling-control common mode is unresolved.
Critical gates	CG-06 and CG-09 UNKNOWN.
Priority action	Separate or independently protect CDU control pathways and produce evidence.
Residual limitation	Recovery evidence remains qualified.
Decision ask	Approve pilot completion; do not claim stronger assurance.

38 Technical report example

Reference implementation requirement.

Element	Pilot specification
Capability	Synthetic distributed training.
Architecture	Power, liquid cooling, compute, storage, network, controls and operations.
Scenarios	Five synthetic scenarios.
Evidence	Twelve synthetic evidence objects.
Controls	Twelve representative local pilot controls.
Findings	Three open findings.
UNKNOWN	Three records, including one critical.
Gates	15 gate results with gate dominance.
Decision	LIMITED.
Traceability	Forward and reverse trace tested.

39 Graph instance summary

Reference implementation requirement.

Element	Pilot specification
CapabilityInstance	Instantiated in synthetic graph
ServiceOutcome	Instantiated in synthetic graph
RequiredFunction	Instantiated in synthetic graph
InfrastructureSystem	Instantiated in synthetic graph
AssetInstance	Instantiated in synthetic graph
Interface	Instantiated in synthetic graph
Dependency	Instantiated in synthetic graph
FaultDomain	Instantiated in synthetic graph
FailureScenario	Instantiated in synthetic graph
Safeguard	Instantiated in synthetic graph
Control	Instantiated in synthetic graph
EvidenceObject	Instantiated in synthetic graph
Assessment	Instantiated in synthetic graph
ControlAssessment	Instantiated in synthetic graph
Finding	Instantiated in synthetic graph
UnknownRecord	Instantiated in synthetic graph
GateResult	Instantiated in synthetic graph
ScoreSnapshot	Instantiated in synthetic graph
AssuranceDecision	Instantiated in synthetic graph
AssuranceReport	Instantiated in synthetic graph
Event	Instantiated in synthetic graph
ReassessmentTrigger	Instantiated in synthetic graph

40 Pilot graph queries

Reference implementation requirement.

Query	Expected semantic answer
Q-01	Return all systems and dependencies supporting SO-04.
Q-02	Return failure paths affecting SO-04.
Q-03	Return evidence supporting CG-06.
Q-04	Return findings and UNKNOWN constraining the decision.
Q-05	Reconstruct executive-report decision text to evidence.
Q-06	Return all objects affected by control-network change.
Q-07	List candidate inferences awaiting review.
Q-08	Return AIDCF-AITG handoff objects only.

41 Reference implementation package

Reference implementation requirement.

Element	Pilot specification
release-manifest.json	Required implementation artifact
known-issues.json	Required implementation artifact
registries/*.csv	Required implementation artifact
schemas/*.json	Required implementation artifact
semantic/aidcf-context.jsonld	Required implementation artifact
semantic/aidcf-core.ttl	Required implementation artifact
semantic/aidcf-shapes.ttl	Required implementation artifact
dataset/capability.jsonld	Required implementation artifact
dataset/architecture.jsonld	Required implementation artifact
dataset/dependencies.jsonld	Required implementation artifact
dataset/scenarios.jsonld	Required implementation artifact
dataset/safeguards.jsonld	Required implementation artifact
dataset/evidence.jsonld	Required implementation artifact
dataset/assessment.jsonld	Required implementation artifact
dataset/gates.jsonld	Required implementation artifact
dataset/score.jsonld	Required implementation artifact
dataset/decision.jsonld	Required implementation artifact
dataset/reports.jsonld	Required implementation artifact
events/change-event.jsonld	Required implementation artifact
tests/valid	Required implementation artifact
tests/invalid	Required implementation artifact
tests/security	Required implementation artifact
tests/migration	Required implementation artifact
conformance-report.json	Required implementation artifact

42 Positive conformance fixtures

Reference implementation requirement.

Fixture	Expected result
POS-01	Capability record valid
POS-02	Outcome/function path valid
POS-03	Architecture snapshot valid
POS-04	Evidence record valid
POS-05	Gate result valid
POS-06	Decision/report consistency valid
POS-07	Event/reassessment valid

43 Negative conformance fixtures

Reference implementation requirement.

Fixture	Expected rejection
NEG-01	Missing capability owner rejected
NEG-02	Broken dependency target rejected
NEG-03	Evidence without assertion link rejected
NEG-04	Gate PASS without evidence rejected
NEG-05	Reasonably Assured with critical UNKNOWN rejected
NEG-06	Report stronger than decision rejected
NEG-07	Supersession cycle rejected
NEG-08	Sensitive export without profile rejected

44 Anti-gaming fixtures

Reference implementation requirement.

ID	Attempt	Expected
AG-01	Convert UNKNOWN to N/A	Reject
AG-02	Duplicate one evidence object to inflate coverage	Detect
AG-03	Average away failed/unknown gate	Reject
AG-04	Remove external dependency from report	Reject
AG-05	Close finding without closure evidence	Reject
AG-06	Promote inferred edge to approved fact	Reject

45 Synthetic change event

Reference implementation requirement.

Element	Pilot specification
eventId	EVT-CHANGE-001
Type	ARCHITECTURE_CHANGED
Subject	Shared cooling control-network path
Synthetic change	CDU B is moved to a separate control-network segment with independent local protection.
Evidence	Updated synthetic topology, configuration and test record.
Initial state	Candidate, not automatically approved.
Trigger	Reassess dependencies, scenarios, controls, CG-06, CG-09, score and decision.
Classification	INTERNAL / SYNTHETIC.

46 Impact-analysis result

Reference implementation requirement.

Element	Pilot specification
Affected dependencies	DEP-05
Affected fault domain	FD-03
Affected scenarios	SCN-02 and SCN-05
Affected safeguard	SG-03, SG-04, SG-09
Affected controls	CTL-004, CTL-009, CTL-010, CTL-011
Affected finding	FND-001
Affected UNKNOWN	UNK-001
Affected gates	CG-06, CG-07, CG-09, CG-10
Affected outputs	Score snapshot, decision, executive report and technical report.

47 Reassessment record

Reference implementation requirement.

Element	Pilot specification
New evidence	EV-013 synthetic independent-control topology; EV-014 synthetic common-mode test.
Disposition	Evidence technically reviewed in pilot.
FND-001	Ready for validation, not auto-closed.
UNK-001	Resolved for synthetic changed snapshot.
CG-06	Candidate PASS after review.
CG-09	Candidate CONDITIONAL due remaining recovery evidence.
Decision effect	Candidate improvement from LIMITED to CONDITIONALLY_ASSURED.
Authority	Requires new decision approval.

48 Superseding decision

Reference implementation requirement.

Element	Pilot specification
New decision	CONDITIONALLY_ASSURED
Conditions	Broaden recovery evidence; confirm OEM support dependency; retain monitoring.
Supersedes	aidcf:decision:synthetic-training-001
Does not erase	Prior limited decision, open history or original evidence.
New snapshot	aidcf:snapshot:synthetic-training-002
Report status	Prior reports superseded; new reports issued.
Restriction	Synthetic demonstration only.

49 AITG handoff registry

Reference implementation requirement.

ID	AIDCF object	AITG view	Mapping
H-01	CapabilityInstance	AITG SystemOfInterest	RELATED
H-02	ServiceOutcome	AITG Outcome	CLOSE
H-03	InfrastructureSystem	AITG InfrastructureObject	CLOSE
H-04	Dependency	AITG Dependency	CLOSE
H-05	FailureScenario	AITG FailureCondition/Path	RELATED
H-06	Safeguard	AITG Control/Breakpoint	RELATED
H-07	EvidenceObject	AITG EvidenceItem	CLOSE
H-08	AssuranceDecision	AITG Decision	CLOSE but federated
H-09	UnknownRecord	AITG UNKNOWN	EXACT within shared profile

50 AITG integration constraints

Reference implementation requirement.

Element	Pilot specification
Identity	Shared real-world identity only when reviewed.
Semantics	No property duplication beyond controlled display/cache.
Evidence	Reuse requires assertion-specific relevance.
Paths	Failure path is not automatically a trust/exposure path.
Decisions	AIDCF and AITG conclusions remain separate and federated.
UNKNOWN	Preserved as first-class state.
Security	Sensitive infrastructure paths restricted.
Automation	No automatic favourable aggregation.

51 Pilot conformance summary

Reference implementation requirement.

Element	Pilot specification
T0	PASS - controlled vocabulary used.
T1	PASS - canonical records and IDs.
T2	PASS - forward/reverse traceability.
T3	PASS for illustrative schema fixtures.
T4	PASS at specification/pilot-model level; independent ontology review pending.
T5	PASS - evidence, gates, decision and reports.
T6	PASS - synthetic change and reassessment.
T7	CONDITIONAL - independent runner and full machine package remain release gates.
Overall	CONDITIONALLY CONFORMANT SYNTHETIC PILOT.

52 Defect and gap register

Reference implementation requirement.

ID	Gap	Disposition
GAP-001	AIDCF-007 control cardinality conflict	Block public baseline
GAP-002	Representative controls use local pilot IDs	Migrate after registry freeze
GAP-003	Synthetic evidence does not establish real effectiveness	Permanent limitation
GAP-004	OWL/SHACL independent review pending	Release gate
GAP-005	No live source connectors exercised	MVP backlog
GAP-006	No production-scale performance test	MVP backlog
GAP-007	Security model specified but not penetration tested	Production gate
GAP-008	Industry-specific regulatory mappings not exercised	Future profiles

53 Lessons learned

Reference implementation requirement.

ID	Pilot lesson
L-01	Baseline consistency is prerequisite to authoritative machine execution.
L-02	Architecture and evidence must be claim-specific.
L-03	UNKNOWN materially changes decisions and cannot be cosmetic.
L-04	Shared control planes create hidden common modes.
L-05	Gate logic prevents favourable scores from masking critical uncertainty.
L-06	Report generation must reference immutable decision snapshots.
L-07	Event handling must separate candidate impact from approved decision change.
L-08	AITG integration works best through reviewed handoffs, not semantic merging.

54 MVP backlog derived from pilot

Reference implementation requirement.

Priority	Capability
P0	Resolve baseline and release manifest
P0	Build canonical registries
P0	Implement JSON schemas and validator
P0	Implement immutable snapshot/lineage store
P0	Implement graph ingest/query
P0	Implement evidence and assessment records
P0	Implement gates and decision constraints
P1	Implement report generation
P1	Implement events and impact analysis
P1	Implement AITG handoff
P1	Build conformance runner
P2	Add source connectors and monitoring
P2	Add security administration and disclosure profiles
P2	Add multi-tenant operations and analytics

55 Repository structure

Reference implementation requirement.

Element	Pilot specification
/docs	AIDCF-015 narrative, pilot report and decisions
/baseline	Release manifest, issues and changelog
/registries	Controlled CSV registries
/schemas	JSON Schemas
/semantic	JSON-LD, Turtle, OWL and SHACL
/dataset	Synthetic capability records
/events	Change and reassessment records
/reports	Executive and technical examples
/tests	Positive, negative, security, migration and anti-gaming
/tools	Validator, loader and query examples
/conformance	Results and release checklist

56 Pilot execution workflow

Reference implementation requirement.

Phase	Activity
P0	Approve charter and synthetic-use boundary
P1	Load baseline and known issues
P2	Instantiate capability/outcomes/architecture
P3	Instantiate dependencies/scenarios/safeguards
P4	Map controls/evidence and assess
P5	Evaluate findings/UNKNOWN/gates/scores
P6	Approve decision and reports
P7	Run validation and graph query suite
P8	Inject change event
P9	Run impact analysis and reassessment
P10	Supersede decision/reports
P11	Publish conformance report and backlog

57 Quality review checklist

Reference implementation requirement.

Element	Pilot specification
Synthetic labels on every output	Required before pilot release
No real vendor/facility claims	Required before pilot release
Baseline issue visible	Required before pilot release
IDs resolve	Required before pilot release
Schemas validate	Required before pilot release
Graph links complete	Required before pilot release
UNKNOWN visible	Required before pilot release
Gate logic verified	Required before pilot release
Score reproducible	Required before pilot release
Reports match decisions	Required before pilot release
Event impact reviewed	Required before pilot release
Prior snapshots retained	Required before pilot release
AITG mappings reviewed	Required before pilot release
Sensitive paths classified	Required before pilot release
Independent reviewer sign-off	Required before pilot release

58 Review and release gates

Reference implementation requirement.

Reviewer/gate	Status
Methodology authority	Pending
Data-centre architect	Pending
Electrical specialist	Pending
Liquid-cooling specialist	Pending
AI/HPC specialist	Pending
Storage/network specialist	Pending
OT/security specialist	Pending
Evidence specialist	Pending
Assurance reviewer	Pending
JSON Schema reviewer	Pending
RDF/OWL/SHACL reviewer	Pending
AITG reviewer	Pending
Security/privacy reviewer	Pending
Independent pilot rerun	Pending
Control registry baseline resolution	Pending
Public release decision	Pending

Final pilot doctrine

AIDCF-015 is a synthetic proof of executability, not a certificate or representation of a real facility. The pilot succeeds only if the complete assurance chain is reproducible in both directions, uncertainty remains visible, gates constrain decisions, reports match approved snapshots, changes trigger reviewed reassessment, and all semantic ownership boundaries remain intact.